The Hubverse: Streamlining Collaborative Infectious Disease Modeling

RSECon26 · Sheffield

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Consortium of Infectious Disease Modeling Hubs

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Background

❌ Where we started

Infectious disease modeling scaled rapidly through COVID-19…

- …but early on the landscape was **fragmented**:
  - Each group used its **own data structures**
  - Results were **hard to integrate or compare**
  - **Modelers and decision-makers** not always aligned

> *"Comparing the accuracy of forecasting applications is difficult because forecasting methods, forecast outcomes, and reported validation metrics varied widely."*
>
> - Chretien et al., PLOS ONE, 2014

✨ The promise of modeling hubs

Modeling hubs coordinate collaborative forecasting:

- Provide **centralised location** for effort coordination
- Define **data standards** and **modeling targets**
- Improve **transparency** and **comparability**
- Aggregate forecasts **enabling ensembles**
- Facilitate timely **public health decision-making**

> *"Collaborative Hubs: Making the Most of Predictive Epidemic Modeling", American Journal of Public Health Reich, et al. 2022*

🕰️ Project origins

- **Pre-COVID:** Forecasting code base existed for CDC **influenza** hubs
- **During COVID:** That code was reused for new **COVID-19** hubs + demand internationally (e.g. Europe) for similar setups
- ❗ Problem: Each hub required **manual editing** of source code

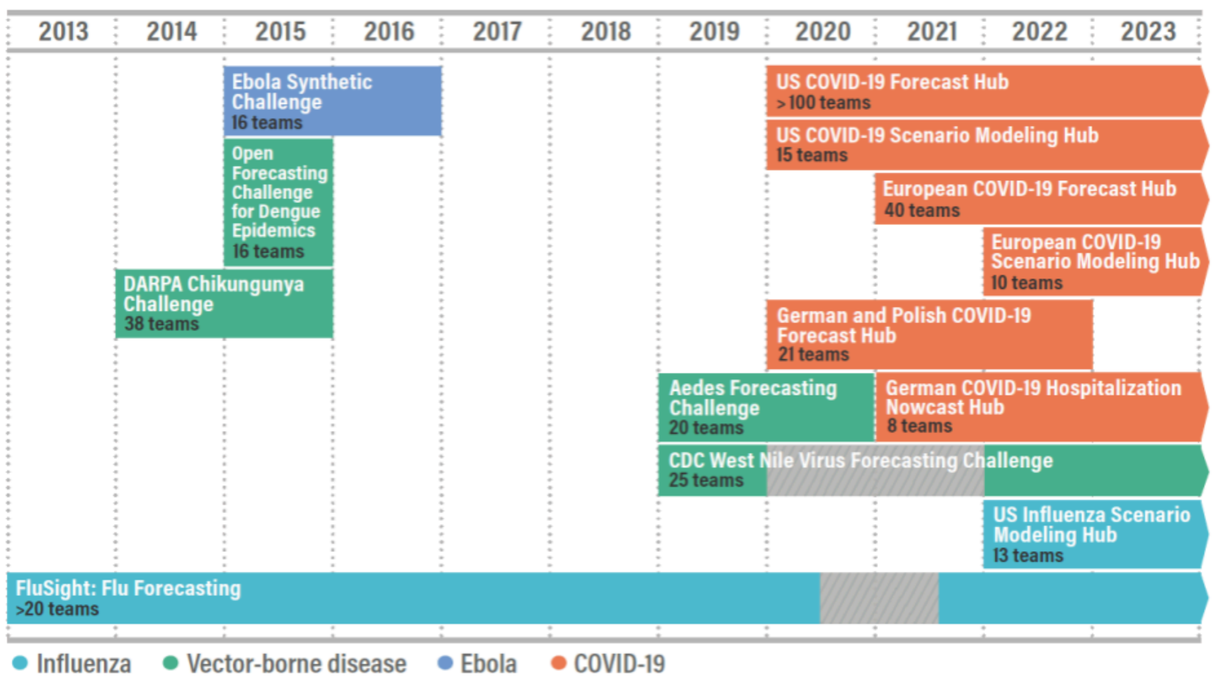


Figure credits: Alex Vespignani and Nicole Samay

🌐 Enter the hubverse

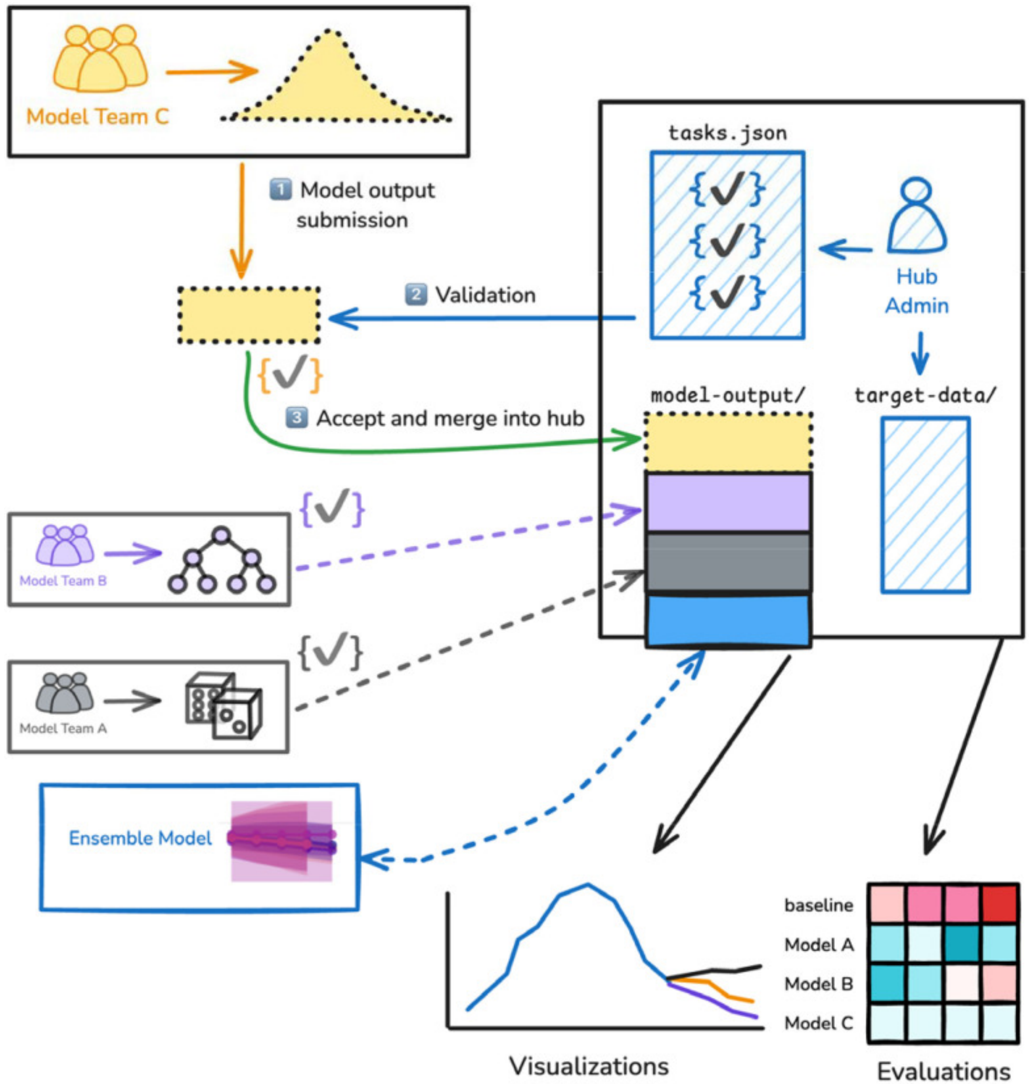
An **open-source software ecosystem** to power modeling hubs:

- GitHub repositories for **centralising hub activity**
- **Data standards** for infectious disease modeling data
- **Schema-driven configuration** for modeling tasks + hub setup
- **Modular tools** for hub administration, data validation, access, evaluation, ensembling and communication

> 📄 **A software platform for collaborative infectious disease modelling (2026)**, *Nature Health, Consortium of Infectious Disease Modeling Hubs et al.* https://doi.org/10.1038/s44360-026-00145-7

Anatomy of a hub

🏗 What a hub actually is



1. Modeling teams **submit model output**
2. Submissions **validated** against `tasks.json`
3. Accepted output **merged into the hub**

Everything downstream, **ensembles**, **visualisations**, **evaluations**, comes for free once the data are standardised.

> Figure 1 from the hubverse paper (preprint, CC BY 4.0)

☑️ A shared data standard

Modeling hubs are **built around a shared data standard:**

- **Structured hub layout**: consistent file system for organizing hub materials
- **Standard model output format**: for file content and naming
- **Modeling task definition**: what’s predicted, by when, for where, in what form

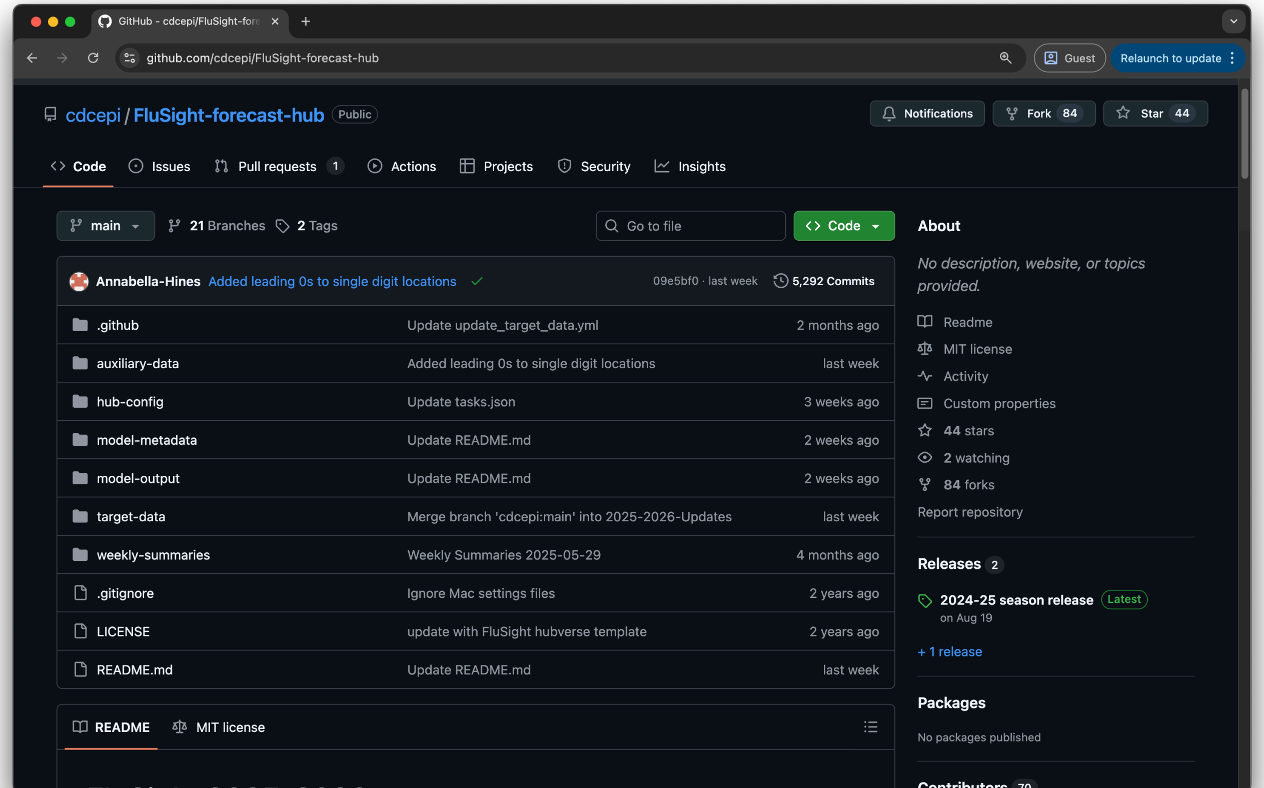
✅ Enable comparability, validation and integration

```
FluSight-forecast-hub/
├── hub-config/
│   ├── admin.json
│   ├── tasks.json
│   ├── model-metadata-schema.json
│   └── target-data.json
├── model-output/
│   └── <model_id>/
│       └── <round_id>-<model_id>.csv
├── model-metadata/
│   └── <model_id>.yml
└── target-data/
```

🦠 Meet the hub: CDC FluSight

The hub we'll follow: CDC FluSight

https://github.com/cdcepi/FluSight-forecast-hub



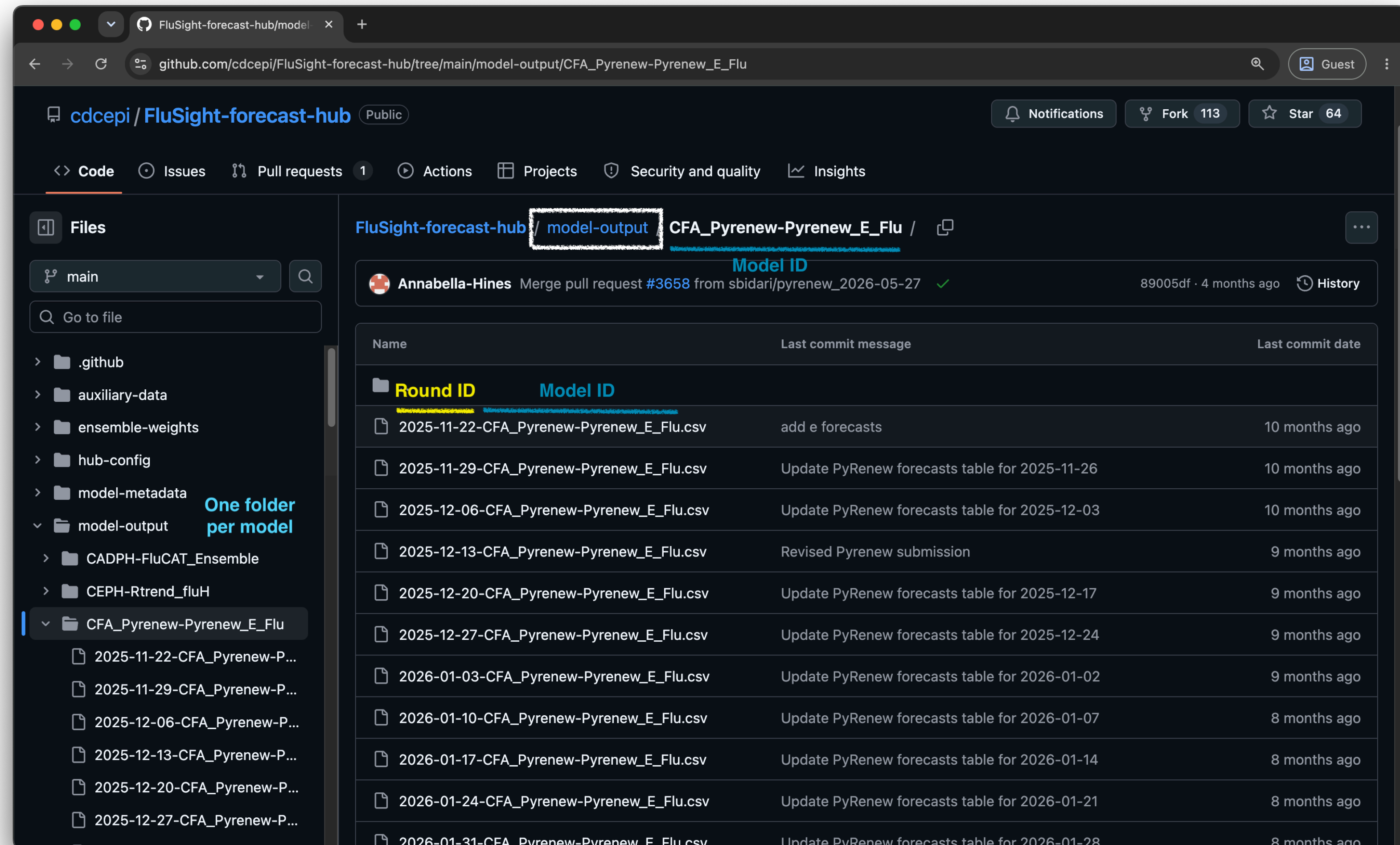
- Used by US CDC to **monitor influenza severity**
- In season, **~60 models** from **~30 teams**, every single week
- Managed with the **full hubverse stack since 2023/24**
- **95 models** from **47 teams** across **88 rounds** since 2023
- Hosted on **GitHub + S3 cloud mirror**

CDC
Centers for Disease Control and Prevention

hubverse

📁 Where model output lives

model-output/<model_id>/<round_id>-<model_id>.csv · one directory per model, one file per round



📘 Model output: expected patterns

Configuring FluSight

⚙️ Config-driven hub setup

Hub administrators configure hubs with structured JSON config files:

- `admin.json`: hub-level metadata
- `tasks.json`: the **scientific question**: what to predict, when, and how
- `model-metadata-schema.json`: what modelers must tell you about their model
- `target-data.json`: what **observed data** should look like

> All validated against a **versioned JSON** schema. The **config** ***is*** **the contract** between administrators, modelers and tooling.

🗃️ `admin.json`: the hub's identity card

```json
{
  "schema_version": ".../schemas/main/v6.0.0/admin-schema.json",
  "name": "US CDC FluSight",
  "maintainer": "US CDC",
  "contact": {
    "name": "Rebecca Borchering",
    "email": "xhq2@cdc.gov"
  },
  "repository": {
    "host": "github",
    "owner": "cdcepi",
    "name": "FluSight-forecast-hub"
  },
  "file_format": ["csv", "parquet"],
  "timezone": "US/Eastern",
  "cloud": {
    "enabled": true,
    "host": {
      "name": "aws",
      "storage_service": "s3",
      "storage_location": "cdcepi-flusight-forecast-hub"
    }
  }
}
```

🎯 Inside `tasks.json`

```
tasks.json
└── rounds[]                                          # a submission cycle
    ├── round_id_from_variable: true                  # round IDs live in the data...
    ├── round_id: "reference_date"                    # ...in this task ID
    ├── submissions_due                               # window, relative to the round
    └── model_tasks[]                                 # groups of related predictions
        ├── task_ids{}                                # the DIMENSIONS of a prediction
        ├── output_type{}                             # the FORM the prediction takes
        └── target_metadata[]                         # what the target means
```

> A hub can have **many rounds**, each with **many model tasks**, so a single hub can ask for **several different kinds of prediction** at once.

Task IDs: the dimensions of a prediction

```json
"task_ids": {
  "reference_date": {
    "required": null,
    "optional": ["2023-10-07", "2023-10-14", "..."]
  },
  "target": {
    "required": null,
    "optional": ["wk inc flu hosp"]
  },
  "horizon": {
    "required": null,
    "optional": [-1, 0, 1, 2, 3]
  },
  "location": {
    "required": null,
    "optional": ["US", "01", "02", "..."]
  },
  "target_end_date": {
    "required": null,
    "optional": ["2023-09-23", "2023-09-30", "..."]
  }
}
```

`reference_date` + `horizon` × 1 week = `target_end_date`

🎯 Targets carry more than a name

`target` is a special task ID: every value in the column gets its own metadata entry

```
"target_metadata": [{
  "target_id": "wk inc flu hosp",
  "target_name": "incident influenza
     hospitalizations",
  "description": "Count of new hospital
     admissions in the week ending...",
  "target_units": "count",
  "is_step_ahead": true,
  "time_unit": "week",
  "target_type": "continuous"
}]
```

Descriptive

- `target_name`, `description`: for humans, and for axis labels

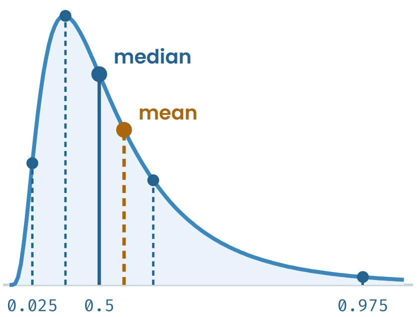
Quantitative

- `target_units`: what is being counted
- `is_step_ahead`, `time_unit`: one step in a sequence
- `target_type`: its statistical type (`continuous`, `ordinal`, `date`, …)

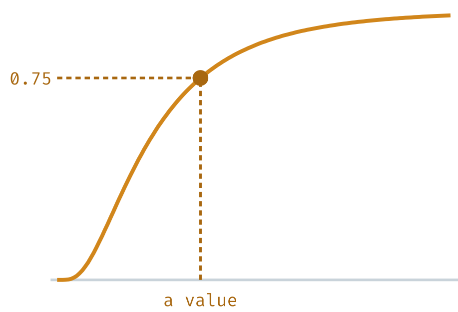
> `target_type` constrains **how a prediction of this target can be expressed**. Which is exactly what an **output type** is…

Output types: how a prediction is expressed

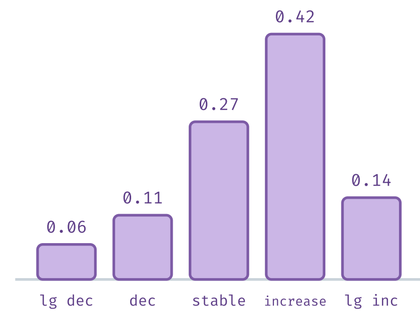
`quantile` · `mean` · `median`
the distribution, and two point estimates



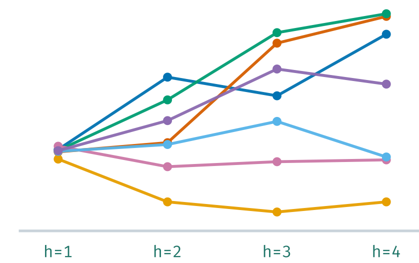
`cdf`
probability at or below a value



`pmf`
probability of each category



`sample`
draws from a joint distribution



output_type	output_type_id	value
mean	*(none)*	mean of the predictive distribution
median	*(none)*	median of the predictive distribution
quantile	a probability level, e.g. `0.75`	the value at that quantile
cdf	a possible value, e.g. `500`	P(outcome ≤ 500)
pmf	a possible category, e.g. `"increase"`	P(outcome = `"increase"`)
sample	a sample index, e.g. `"s3"`	one draw from the distribution

…and how a hub configures them

```json
"output_type": {
  "quantile": {
    "output_type_id": {
      "required": [0.01, 0.025, "...", 0.975, 0.99]
    },
    "is_required": true,
    "value": { "type": "double", "minimum": 0 }
  },
  "sample": {
    "output_type_id_params": {
      "compound_taskid_set": ["reference_date", "location", "target"],
      "min_samples_per_task": 100,
      "max_samples_per_task": 100
    },
    "is_required": false,
    "value": { "type": "integer", "minimum": 0 }
  }
}
```

🔍 Configured model output in the wild

Real FluSight rows from round `2026-01-10`, national. **task IDs** · **output type** · **value**

reference_date	horizon	target	target_end_date	location	output_type	output_type_id	value
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.01	18429
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.025	20177
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.05	21981
						…	
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.5	38936
						…	
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.975	61292
2026-01-10	1	wk inc flu hosp	2026-01-17	US	quantile	0.99	68125
2026-01-10	0	wk inc flu hosp	2026-01-10	US	sample	us_s1	36268
2026-01-10	1	wk inc flu hosp	2026-01-17	US	sample	us_s1	36397
2026-01-10	2	wk inc flu hosp	2026-01-24	US	sample	us_s1	33669
2026-01-10	3	wk inc flu hosp	2026-01-31	US	sample	us_s1	33727
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	large_decrease	0.114
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	decrease	0.244
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	stable	0.158
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	increase	0.231
2026-01-10	1	wk flu hosp rate change	2026-01-17	US	pmf	large_increase	0.253

Hubs collect metadata about the models too

`model-metadata-schema.json` sets out what each team must document about their model

- One file per model in `model-metadata/`, validated on every PR
- The hubverse ships a schema template, hubs adapt it

```yaml
team_name: "UMass-Amherst"
model_abbr: "flusion"
model_version: "1.1"
model_contributors:
  - name: "Evan Ray"
    affiliation: "UMass Amherst"
license: "CC-BY-4.0"
data_inputs: "NHSN, FluSurv-NET and ILINet."
methods: "Ensemble of statistical and machine
  learning time series models."
designated_model: true
```

> Provenance for every prediction, and a natural basis for **model cards**.

Beyond the data standard

🔁 Automating everything we can

GitHub Actions do the operational work:

- ✅ PR-level **model output validation**
- ✅ **Hub configuration** validation
- ☁️ **Cloud** hub data syncing
- 📊 **Dashboard builds**

> All actions live in **`hubverse-actions`** and install with `hubCI::use_hub_github_action()`

☁️ Cloud storage and access

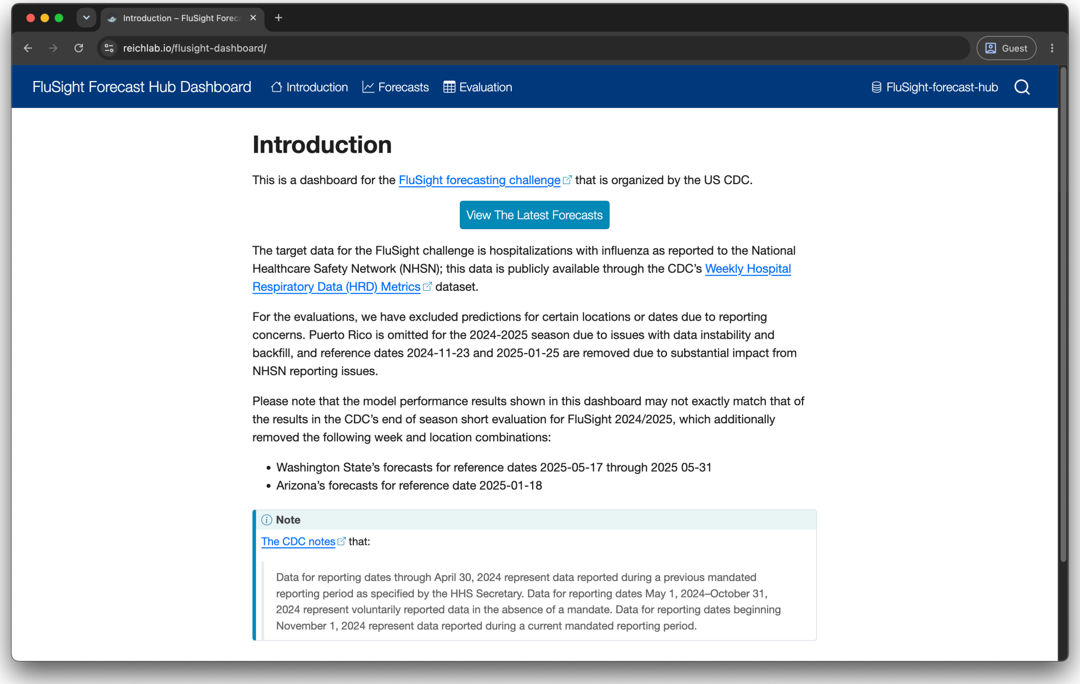
Validated hub data is mirrored to a public S3 bucket

- Opened directly as an **Arrow dataset**: no cloning, no full download
- **Query-able** via 📦 `hubData` (R) and 📦 `hubdata` (Python)
- Turns a hub into an **open data resource**, not just a repository

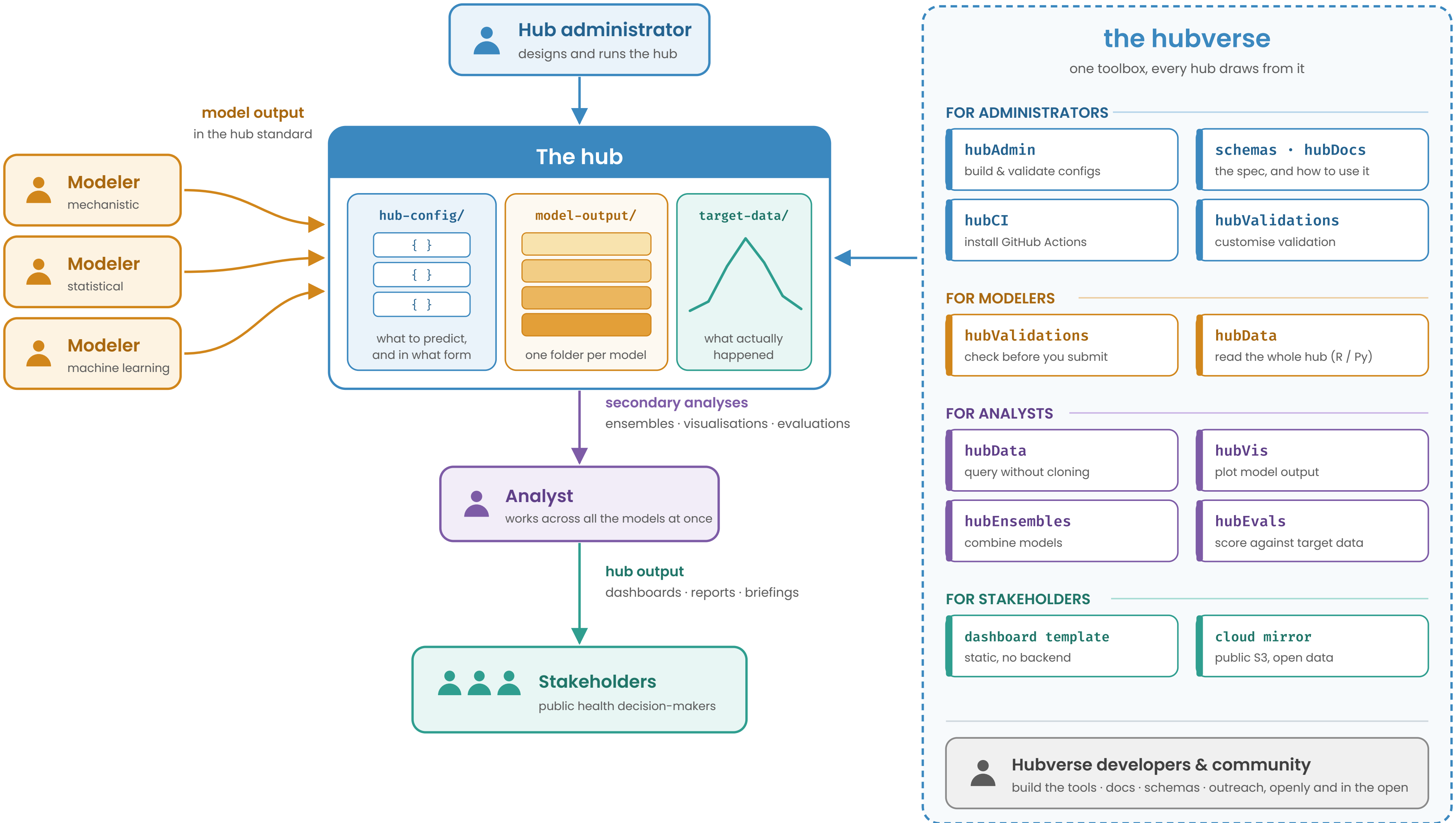
> Enable it with a few lines in `admin.json`. The hubverse provisions the bucket.

📊 Dashboards & communication

- Built with **Quarto** so easily customisable via Quarto configuration
- Deployed as a **fully static site**, no backend required
- Powered by JSON data prepared via GitHub workflows
- Interactive UI built with client-side JavaScript (fast!)
- New instances set up by copying/configuring the `hub-dashboard-template`



👥 Hubverse roles and their tooling

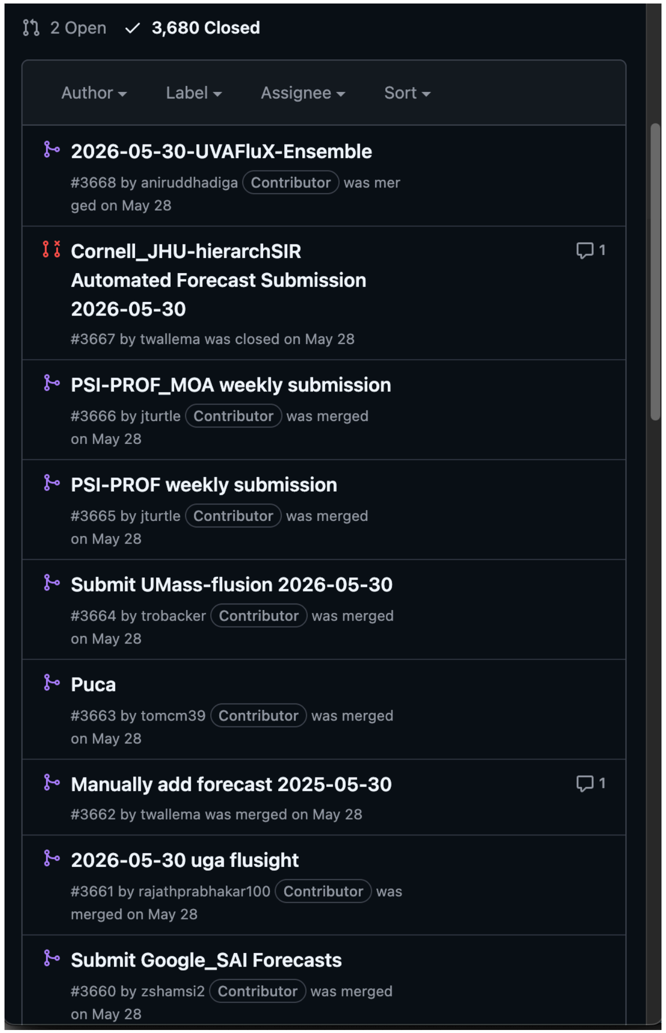


hubverse

🦠 FluSight in operation

✅ Model output validation with `hubValidations`

Submitted by pull request, validated by GitHub Actions, **reported back on the PR**



Passed | Failed

github-actions bot commented

Submission validation

✅ **All validation checks passed.**

▸ Check results

Results for commit `a1b2c3d`

🔎 What actually gets checked

Standard checks, generated from `tasks.json`

- 📄 **File**: name, location, format, submission window
- 🧱 **Structure**: columns, types, duplicates, round ID
- 🎯 **Content**: valid task ID combinations, required tasks present, values in bounds
- 📐 **Output-type specific**: quantiles don't cross, `pmf` sums to 1, sample counts
- 🏷️ **Model metadata** and 🎯 **target data**: each with their own checks

Hub-specific checks in `hub-config/validations.yml`

```
default:
  validate_model_data:
    horizon_timediff:
      fn: "opt_check_tbl_horizon_timediff"
      pkg: "hubValidations"
      args:
        t0_colname: "reference_date"
        t1_colname: "target_end_date"
```

- **Opt-in** checks shipped with `hubValidations`
- Or the hub's **own R functions**, in `src/validations/R`

> `create_custom_check()` scaffolds a custom check: right structure, return classes and conventions, ready to fill in.

📂 Accessing model output via `hubData`

Connect to Arrow dataset of forecast submissions

```r
library(hubData)

hub_path <- s3_bucket(
  "cdcepi-flusight-forecast-hub"
)
hub_con <- connect_hub(hub_path)
hub_con
```

```
hub_connection
9 columns
reference_date: date32[day]
target: string
horizon: int32
target_end_date: date32[day]
location: string
output_type: string
output_type_id: string
value: double
model_id: string
```

Query and collect data

```r
# Filter for one model and forecast date using dplyr
library(dplyr)
hub_con |>
  filter(
    model_id == "CADPH-FluCAT_Ensemble",
    target_end_date == "2023-10-28"
  ) |>
  collect_hub()
```

```
# A tibble: 92 × 9
  model_id     reference_date target horizon target_end_date location output_type
* <chr>        <date>         <chr>    <int> <date>          <chr>    <chr>
1 CADPH-FluC… 2023-10-14     wk in…       2 2023-10-28      06       quantile
2 CADPH-FluC… 2023-10-14     wk in…       2 2023-10-28      06       quantile
3 CADPH-FluC… 2023-10-14     wk in…       2 2023-10-28      06       quantile
4 CADPH-FluC… 2023-10-14     wk in…       2 2023-10-28      06       quantile
5 CADPH-FluC… 2023-10-14     wk in…       2 2023-10-28      06       quantile
# ℹ 87 more rows
# ℹ 2 more variables: output_type_id <chr>, value <dbl>
```

See more in Accessing data vignette.

Python analogue **`hub-data`** also available.

🌐 Ensembling with `hubEnsembles`

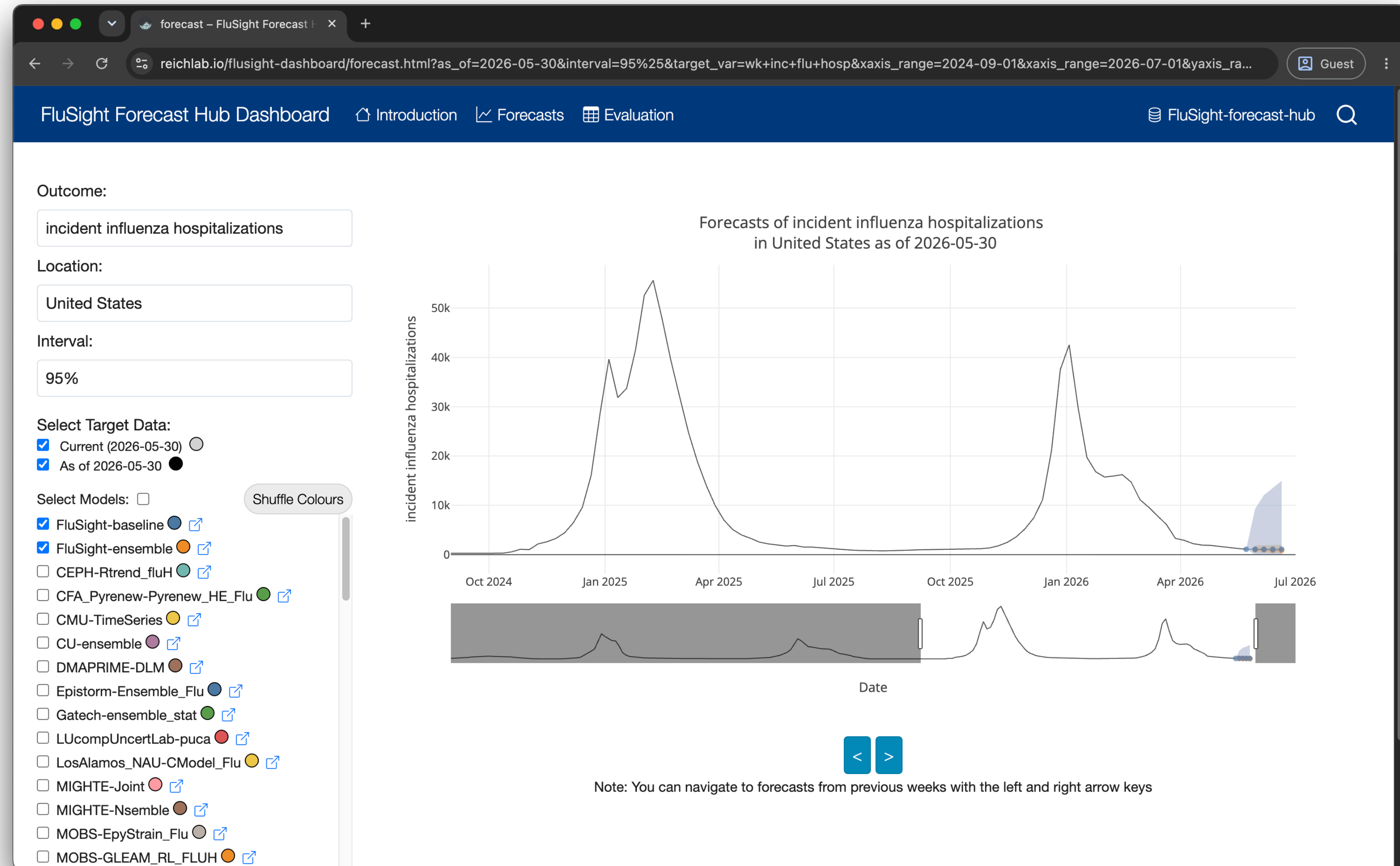
Combine models using simple or weighted rules

```r
forecast_df <- hub_con |>
  filter(
    model_id %in%
      c(
        "CADPH-FluCAT_Ensemble",
        "CEPH-Rtrend_fluH",
        "CFA_Pyrenew-Pyrenew_HE_Flu"
      ),
    output_type == "quantile"
  ) |>
  collect_hub()


hubEnsembles::simple_ensemble(
  forecast_df,
  agg_fun = median,
  model_id = "simple-ensemble-median"
)
```

```
# A tibble: 492,476 × 9
  model_id    reference_date target horizon target_end_date location output_type
* <chr>       <date>         <chr>    <int> <date>          <chr>    <chr>
1 simple-ens… 2023-10-14     wk in…      -1 2023-10-07      01       quantile
2 simple-ens… 2023-10-14     wk in…      -1 2023-10-07      01       quantile
3 simple-ens… 2023-10-14     wk in…      -1 2023-10-07      01       quantile
4 simple-ens… 2023-10-14     wk in…      -1 2023-10-07      01       quantile
5 simple-ens… 2023-10-14     wk in…      -1 2023-10-07      01       quantile
# ℹ 492,471 more rows
# ℹ 2 more variables: output_type_id <chr>, value <dbl>
```

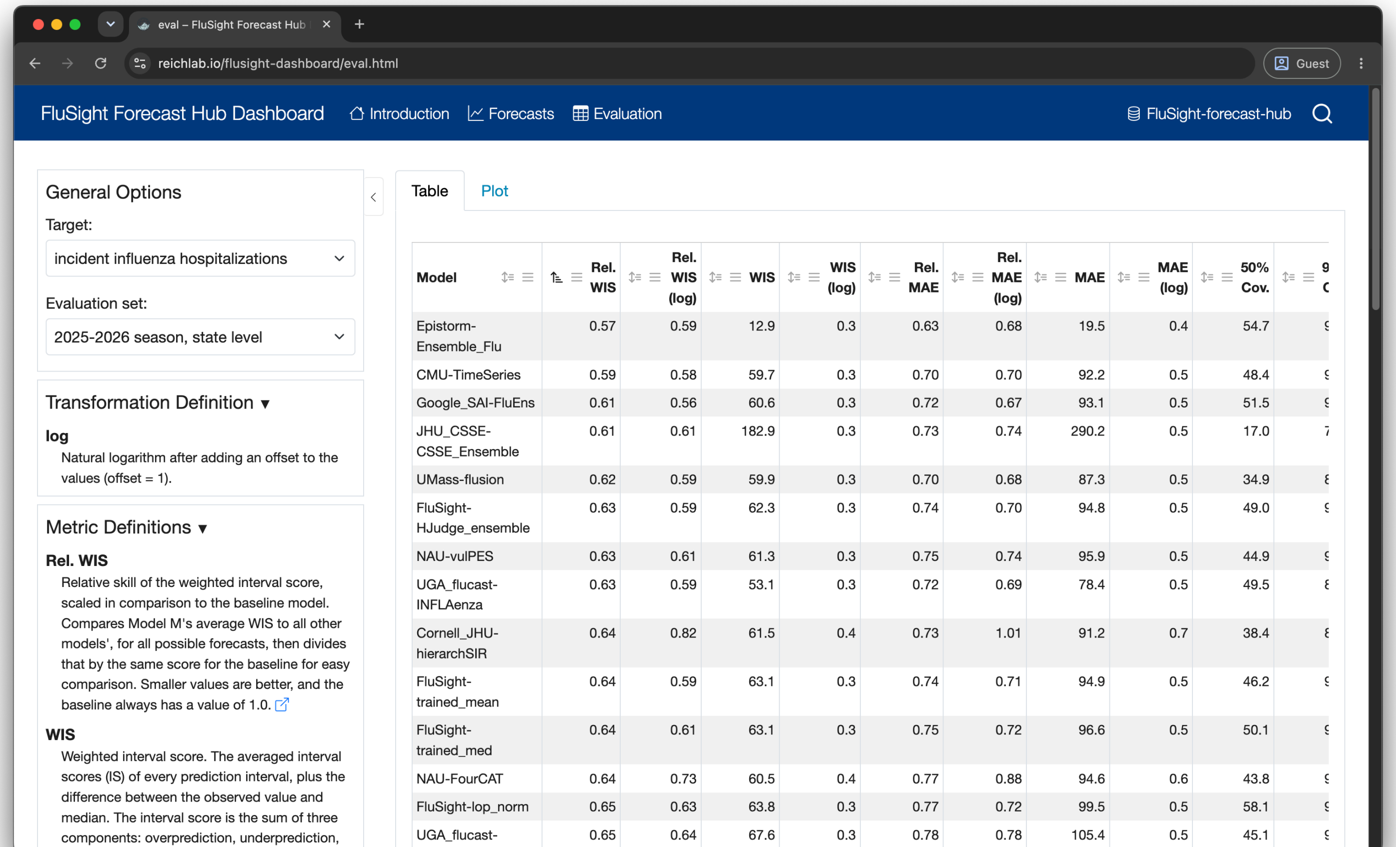
📈 Dashboard - forecasts



Dashboard - model evaluations

Evaluates forecasts against target (observed) data.

Table Plot



Model	Rel. WIS	Rel. WIS (log)	WIS	WIS (log)	Rel. MAE	Rel. MAE (log)	MAE	MAE (log)	50% Cov.
Epistorm-Ensemble_Flu	0.57	0.59	12.9	0.3	0.63	0.68	19.5	0.4	54.7
CMU-TimeSeries	0.59	0.58	59.7	0.3	0.70	0.70	92.2	0.5	48.4
Google_SAI-FluEns	0.61	0.56	60.6	0.3	0.72	0.67	93.1	0.5	51.5
JHU_CSSE-CSSE_Ensemble	0.61	0.61	182.9	0.3	0.73	0.74	290.2	0.5	17.0
UMass-flusion	0.62	0.59	59.9	0.3	0.70	0.68	87.3	0.5	34.9
FluSight-HJudge_ensemble	0.63	0.59	62.3	0.3	0.74	0.70	94.8	0.5	49.0
NAU-vulPES	0.63	0.61	61.3	0.3	0.75	0.74	95.9	0.5	44.9
UGA_flucast-INFLAenza	0.63	0.59	53.1	0.3	0.72	0.69	78.4	0.5	49.5
Cornell_JHU-hierarchSIR	0.64	0.82	61.5	0.4	0.73	1.01	91.2	0.7	38.4
FluSight-trained_mean	0.64	0.59	63.1	0.3	0.74	0.71	94.9	0.5	46.2
FluSight-trained_med	0.64	0.61	63.1	0.3	0.75	0.72	96.6	0.5	50.1
NAU-FourCAT	0.64	0.73	60.5	0.4	0.77	0.88	94.6	0.6	43.8
FluSight-lop_norm	0.65	0.63	63.8	0.3	0.77	0.72	99.5	0.5	58.1
UGA_flucast-	0.65	0.64	67.6	0.3	0.78	0.78	105.4	0.5	45.1

Closing remarks

🪩 Who's using it

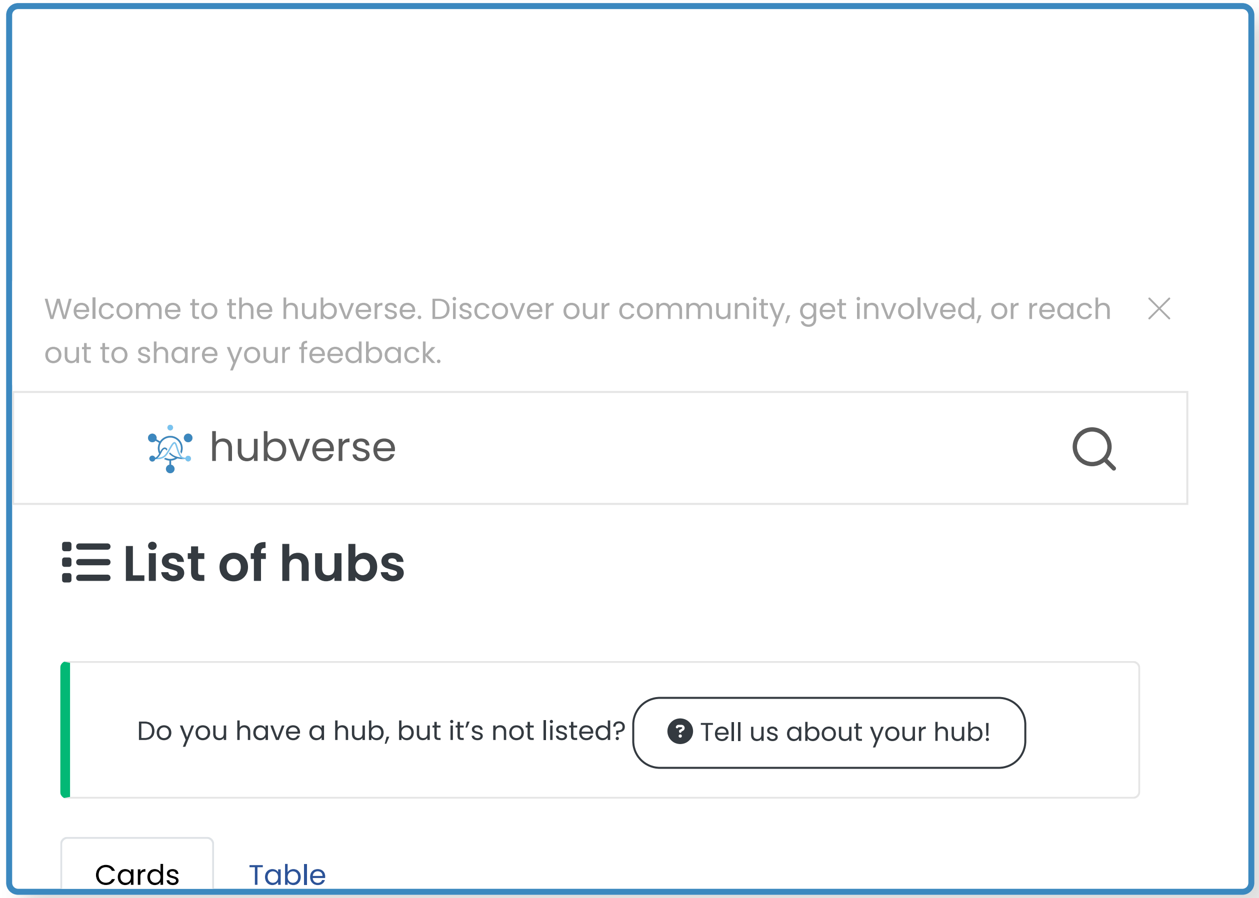
34 hubs across **15 organizations**, on five continents

736 models · **4.6 billion rows** of model output *(Sep 2026)*

Five kinds of hub:

- 🌍 **Community**: open, public repos
- 🔒 **Collaborative, private**: e.g. Paraguay, Australia–Aotearoa
- 🧪 **Local / model development**: a hub on your own laptop
- 🎓 **Training**: teaching hubs
- 🗄️ **Archival**: retired hubs, kept queryable

🔗 hubverse.io/community/hubs.html



💡 Lessons & wider relevance

- 📐 **A good standard is leverage**: define the interface once, reuse everything downstream
- ✅ **Standards + automation reduce friction**
- ⚙️ **Configuration over code**: versioned schemas let the standard evolve without breaking hubs
- 🧰 **Open source keeps it free & accessible**
- 🌍 **Standardised, open data fuels downstream use**: teaching, reproducible research, and teams' own model development

Built on hubverse data

SISMID 🔗

Short course teaching real-world outbreak forecast evaluation, worked directly on hub data

RespiLens 🔗

A responsive web app to visualise US respiratory disease forecasts, built for state health departments and the public. ACCIDDA, UNC Chapel Hill

MicroHub workshop 🔗

Hands-on workshop: run forecast models locally in a Docker app, then stand up your own hubverse hub on GitHub. Northern Arizona University

🙏 Thank you!

- 🔗 https://hubverse.io
- 📘 https://docs.hubverse.io
- hubverse-org
- 📦 **Install the packages:** https://hubverse-org.r-universe.dev
- ✉ info@r-rse.eu

Members of hubverse team at the 2024 work retreat, UMass Amherst

Tip

Interested in getting involved? Check out our **Getting Involved** page!

Read more

Consortium of Infectious Disease Modeling Hubs *et al.* (2026)
A software platform for collaborative infectious disease modelling
Nature Health

https://doi.org/10.1038/s44360-026-00145-7

Open-access preprint: https://doi.org/10.1101/2025.10.03.25337284